KiMO: Knowledge-infused Multi-agent Orchestrator

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ABSTRACT

Multi-Agent Systems (MAS) provide a natural paradigm for solving complex problems requiring coordinated, multi-step reasoning and decision making. As these systems scale, effective agentic orchestration becomes crucial. While existing tools such as AgentFlow and Agent Development Kit (ADK) support agent communication and task delegation, they remain limited in understanding task semantics, coordinating heterogeneous agent types (e.g., reactive vs. cognitive), and aligning agent selection with domain knowledge and infrastructure constraints. We present *KiMO*, a knowledge-infused multi-agent orchestrator that makes coordination explicit through structured knowledge infusion: *planning ontologies* encode domain-specific task decompositions, and *agent registries* formalize capabilities and compatibility constraints for heterogeneous components. KiMO operationalizes these through a two-stage pipeline: ontology-guided sub-task planning (PlanGen) and registry-guided workflow assembly (AgentGen). This dual knowledge infusion enables interpretable, flexible orchestration with human-inspectable coordination decisions. Our interactive platform demonstrates real-time workflow construction for manufacturing pipelines with expert refinement and dynamic knowledge extension. Code is available at https://tinyurl.com/3es9zyhj and Demo Video at https://tinyurl.com/yfa8ctwn.

KEYWORDS

Multi-agent orchestration, Knowledge Infusion, Workflow construction, Heterogeneous agents

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1 INTRODUCTION AND RELATED WORK

Large Language Models (LLMs) have enabled automation beyond static rule execution, leading to increased interest in agent-driven workflow systems [1, 3, 12]. This has driven a shift from traditional RPA towards *Agentic Process Automation* (APA) [9, 13], where workflows are dynamically LLM-guided. However, heavy reliance on LLMs without explicit task semantics leads to unreliable orchestration in safety-critical domains. Recent multi-agent systems (MAS) [7, 8, 11, 14] distribute roles across collaborating agents, but orchestration remains implicit and tightly coupled to LLM-centric agents, with limited support for heterogeneous components (reactive agents, statistical models, symbolic processes). While progress has enhanced individual agent reasoning [2, 4], the coordination layer governing task decomposition and agent interaction remains to be a bottleneck for reliability and human oversight as MAS scale.

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Why existing tools fall short: AgentFlow [10] and AutoGen [14] provide agent communication but lack semantic task understanding and heterogeneous agent coordination. ADK [6] supports delegation but assumes homogeneous LLM agents, limiting integration of specialized non-LLM components.

Our approach: We present KiMO, which makes orchestration explicit through *Knowledge Infusion* (KI) [5]. KI is a process that infuses domain expert knowledge into stochastic models for domain-grounded expertise. KiMO applies KI at two levels: (1) *planning ontologies* encode hierarchical task decompositions from expert workflows, and (2) *agent registries* formalize capabilities and I/O specifications for heterogeneous components. This dual knowledge infusion enables KiMO's two-stage pipeline: *PlanGen* generates ontology-grounded sub-task plans, then *AgentGen* assembles registry-grounded workflows with capability-matched agents. This approach supports flexible orchestration of heterogeneous agents (e.g. LLM-based, statistical, symbolic) with human-inspectable coordination decisions (Figure 1).

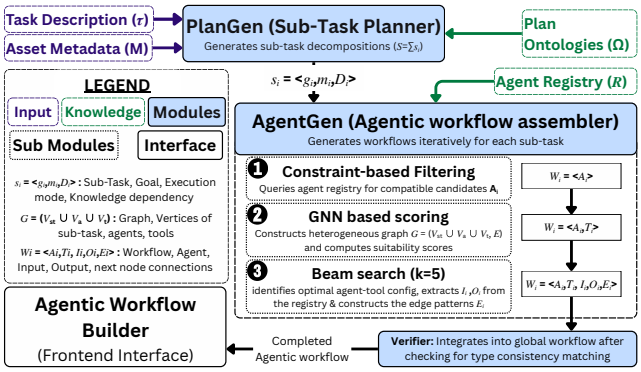

Figure 1: System Architecture

2 SYSTEM OVERVIEW

KiMO implements a two-stage pipeline infusing structured knowledge at each stage: planning ontologies (Ω) guide task decomposition, while agent registry ($\mathcal{R}$) constrains workflow assembly (Figure 1).

Stage 1: Ontology-guided Sub-task Planning (PlanGen). Given task description τ and metadata $\mathcal{M}$ (Figure 2A), PlanGen generates sub-task decompositions $P = \{s_1, \ldots, s_n\}$ where $s_i = \langle g_i, m_i, D_i \rangle$ specifies goal, execution mode (sequential/parallel), and knowledge dependencies. PlanGen employs a fine-tuned LLM (Llama 3.2 3B with LoRA) conditioned on plan ontologies (Ω) via retrieval augmented fine-tuning (RAFT) [15]. *Example:* For τ="detect manufacturing defects", ontology Ω containing hierarchical task structures yields s_1=$\langle$collect sensor data, sequential, $\emptyset\rangle$, s_2=$\langle$preprocess signals, sequential, $D_1\rangle$, s_3=$\langle$run CNN inference, parallel, $D_2\rangle$. The resulting decompositions P are rendered on an interactive canvas (Figure 2B) where experts can modify goals, dependencies, or execution modes.

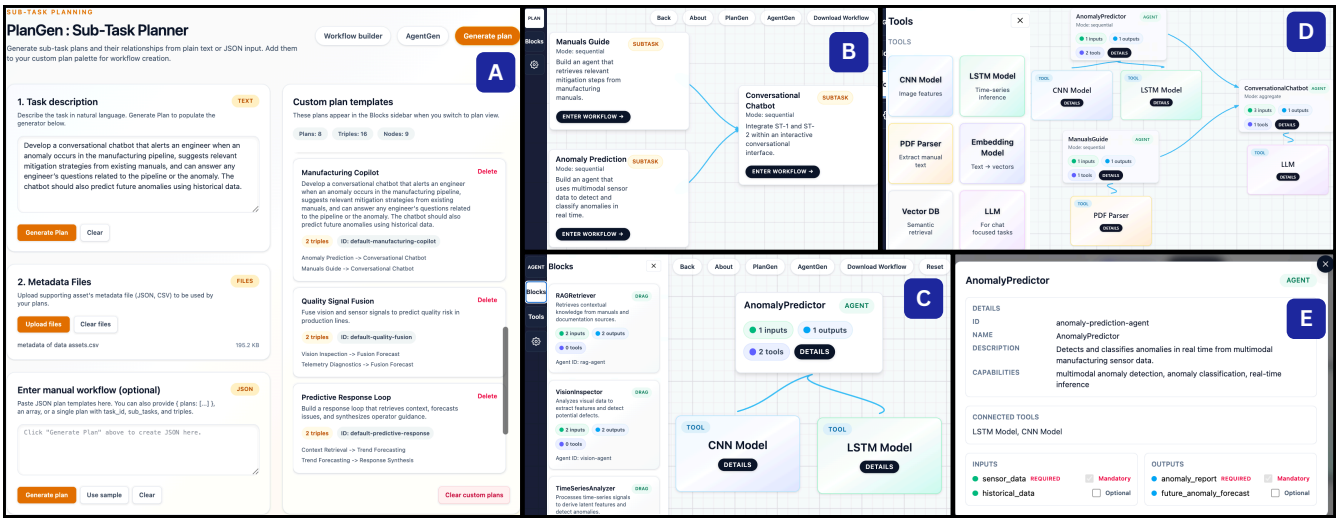

Figure 2: Real-time workflow construction for manufacturing anomaly detection: (A) PlanGen interface accepts task description and metadata; (B) Generated sub-task plan with dependencies; (C) AgentGen constructs agentic workflow for anomaly prediction sub-task, showing selected agent (AnomalyPredictor) and connected tools; (D) Complete manufacturing pipeline in workflow builder canvas with all agents and data streams; (E) Agent specification panel showing capabilities, I/O schemas, and tool connections. Note: Users can modify at any stage as changes propagate through Verifier before integration.

Stage 2: Registry-guided Workflow Assembly (AgentGen). For each sub-task s_i received from PlanGen, AgentGen iteratively constructs workflow $w_i = \langle A_i, T_i, I_i, O_i, E_i \rangle$ specifying agents, tools, input/output data streams, and inter-agent connections through three registry-guided phases (Figure 1): **(i) Constraint-based Filtering** queries agent registry $\mathcal{R}$ for compatible candidates $A_{\text{cand}} = \{a \in \mathcal{R} \mid \text{capability}(a) \cap \text{required}(s_i) \neq \emptyset\}$ based on sub-task goals, yielding $w_i = \langle A_i \rangle$. **(ii) GNN-based Scoring** constructs heterogeneous graph $G = (V_{\text{subtask}} \cup V_{\text{agents}} \cup V_{\text{tools}}, E)$ over sub-tasks, agents, and tools, computing suitability scores via message passing over compatibility relations and historical profiles from $\mathcal{R}$, producing $w_i = \langle A_i, T_i \rangle$. *Example:* For s_3="run CNN inference", registry $\mathcal{R}$ contains AnomalyPredictor (type: cognitive, capability: CNN/LSTM, input: tensor, GPU: required) and StatisticalDetector (type: reactive, capability: threshold, input: scalar, GPU: false); GNN scores AnomalyPredictor=0.89 (capability match + GPU available) vs StatisticalDetector=0.34. **(iii) Beam Search** (width $k = 5$) identifies optimal agent-tool configurations and extracts I/O specifications I_i, O_i and constructs connection patterns E_i from registry metadata, completing $\langle A_i, T_i, I_i, O_i, E_i \rangle$ for a single sub-task (Figure 2C). A Verifier validates type consistency before integrating into the global workflow displayed in the Agentic Workflow Builder interface (Figure 2D). Executed workflows are logged to retrain the GNN, enabling continuous improvement.

3 DEMONSTRATION PLATFORM

Our web-based platform enables interactive construction and execution of knowledge-infused multi-agent workflows through two integrated views (Figure 2).

Planning View (A,B): Users input task description τ and metadata $\mathcal{M}$ to invoke PlanGen's ontology-guided decomposition. Generated sub-tasks render as an interactive graph displaying execution modes and dependencies D_i. The interface supports real-time modification of decompositions and loading of custom templates, enabling domain experts to extend ontology Ω dynamically.

Agentic Workflow View (C,D,E): Selecting any sub-task s_i triggers AgentGen's three-phase assembly with live visualization of constraint filtering, GNN scoring, and beam search. Panel C displays the generated workflow graph with agent nodes, tool connections, and data streams for the Anomaly Prediction sub-task from the manufacturing pipeline. Panel E reveals detailed agent specifications from registry $\mathcal{R}$, including the AnomalyPredictor's capabilities, I/O schemas, and connected tools. Users can modify workflows or construct custom agents by selecting tools from registry $\mathcal{R}$ (Panel D left sidebar) and defining I/O specifications. The complete manufacturing pipeline workflow with all integrated agents and data streams is visualized in Panel D.

Interactive Capabilities: The platform supports custom plan creation through PlanGen, custom agent construction through AgentGen by composing registry tools, real-time workflow modification with type-consistency verification, and pipeline execution with logged outputs. All executed workflows are retained to enable GNN retraining and continuous system improvement.

4 CONCLUSION AND FUTURE WORK

KiMO demonstrates that explicit knowledge infusion through planning ontologies and agent registries enables reliable orchestration of heterogeneous multi-agent systems. The two-stage architecture supports interpretable coordination and dynamic knowledge extension. Future work includes comprehensive workflow quality metrics, federated registries for cross-organization agent sharing, and enhanced continual learning mechanisms.

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REQUIREMENTS FOR DEMO SETTING AT THE CONFERENCE

To facilitate the demonstration of KiMO at the conference, the following requirements are necessary:

- **Hardware:**
  - A laptop or desktop computer with at least 16GB RAM, a modern multi-core CPU, and a GPU with a minimum of 6GB VRAM (e.g., NVIDIA RTX 3060 or equivalent).
  - External monitor to display PlanGen and AgentGen in action and to showcase the execution of bash scripts of the constructed workflow.
  - A projector or large display screen for audience visualization of Agent builder whiteboard interface.
- **Networking:**
  - Stable internet connectivity.
  - Access to the conference network for remote access to supplementary data, if needed.
- **Software and Environment:**
  - Pre-installed Python environment (version 3.8 or above) with necessary dependencies such as PyTorch, Transformers, and OpenCV.
  - Docker installed on the machine to deploy containerized instances of SmartPilot agents.
- **Other Equipment:**
  - A wireless presenter or remote for interactive demonstrations.
  - A set of printed handouts or a QR code linking to supplementary materials such as the video demo, datasets, and documentation.
- **Space:**
  - A table or booth with sufficient space for hardware setup and audience interaction.
  - Power outlets with surge protection for all devices.

These requirements are essential to ensure a smooth and interactive demonstration of KiMO's capabilities at the conference.

SUPERVISOR ENDORSEMENT

Amit Sheth,
NCR Chair & Professor.
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Dear AAMAS Demo Reviewers,

I am writing to describe the student contributions to the project **"KiMO: Knowledge infused Multiagent Orchestrator,"** which has been submitted to the **AAMAS Demo track.**

Ritvik Garimella led the project and played a pivotal role in its success. He was responsible for the conceptualization and methodology development, ensuring the project was grounded in a solid theoretical framework. He actively contributed to software development, validation, formal analysis, ontology construction and data curation. Ritvik was involved in drafting the manuscript, creating visualizations, performing review and editing tasks. He also supervised the team and managed the overall administration of the project.

Michael Stewart's role was instrumental in shaping the project's conceptualization and methodology. He contributed to software development, validation, formal analysis, and data curation. Michael worked closely on the software development and taking core UI design decisions.

Kevin Do contributed to the software development aspects of the project. He participated in the validation, formal analysis, and data curation, ensuring the quality and integrity of the results alongside Michael.

Leeon Israel focused on software development and validation tasks, ensuring the technical implementation met the project's requirements. He also contributed to creating visualizations that highlighted the project's key findings.

Each student played a critical role in the project's development, demonstrating collaboration, technical expertise, and dedication to its success.

Sincerely,

Amit P. Sheth

01/09/2026

Amit Sheth
NCR Chair & Professor.
Founding Director,
AI Institute;
University of South Carolina.

Date